

Widening

CSE552 Program Analysis — Lecture 9

Jaemin Hong

Interval Analysis

- ▶ An interval analysis computes a lower and an upper bound for the possible values of each integer variable
- ▶ Sound answers can be used for optimizations and bug detection related to array bounds checking, numerical overflows, and integer representations
- ▶ Involves a lattice of infinite height

Interval Domain — Definitions

$$\overline{\mathbb{Z}} = \mathbb{Z} \cup \{-\infty, \infty\}$$

$$\text{Interval} = \{[l, h] \mid l, h \in \overline{\mathbb{Z}} \wedge l \leq h\} \cup \{\perp\}$$

$$[l_1, h_1] \sqsubseteq [l_2, h_2] \iff l_2 \leq l_1 \wedge h_1 \leq h_2$$

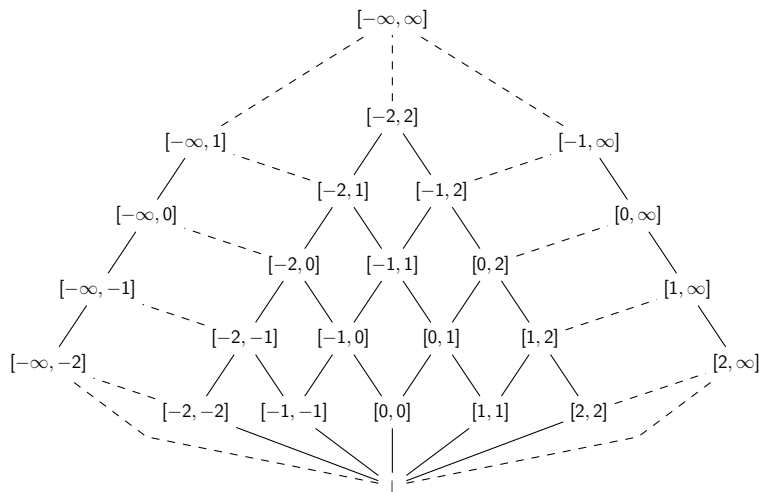
$$\top = [-\infty, \infty]$$

Interval Domain — Join and Meet

$$[l_1, h_1] \sqcup [l_2, h_2] = [\min(l_1, l_2), \max(h_1, h_2)]$$

$$[l_1, h_1] \sqcap [l_2, h_2] = \begin{cases} [\max(l_1, l_2), \min(h_1, h_2)] & \text{if } \max(l_1, l_2) \leq \min(h_1, h_2) \\ \perp & \text{otherwise} \end{cases}$$

Interval Domain — Hasse Diagram



Interval Domain — Infinite Height

$$[0, 0] \sqsubseteq [0, 1] \sqsubseteq [0, 2] \sqsubseteq [0, 3] \sqsubseteq [0, 4] \sqsubseteq [0, 5] \sqsubseteq \dots$$

Transfer Functions

State = Var $\rightarrow$ Interval

$x = e :$ $t_v(\sigma) = \sigma[x \mapsto \text{eval}(\sigma, e)]$

$\text{if } x :$ $t_v(\sigma) = \sigma$

$\text{return} :$ $t_v(\sigma) = \sigma$

$\sigma_{\text{start}} = \top$

Abstract Evaluation

$$\text{eval}(\sigma, x) = \sigma(x)$$

$$\text{eval}(\sigma, n) = [n, n]$$

$$\text{eval}(\sigma, \text{input}()) = \top = [-\infty, \infty]$$

$$\text{eval}(\sigma, e_1 \text{ op } e_2) = \widehat{op}(\text{eval}(\sigma, e_1), \text{eval}(\sigma, e_2))$$

Abstract Operations

$$\widehat{op}([l_1, h_1], [l_2, h_2]) = \left[\min_{\substack{x \in [l_1, h_1] \\ y \in [l_2, h_2]}} x \text{ op } y, \max_{\substack{x \in [l_1, h_1] \\ y \in [l_2, h_2]}} x \text{ op } y \right]$$

$$\widehat{+}([l_1, h_1], [l_2, h_2]) = [l_1 + l_2, h_1 + h_2]$$

$$\widehat{-}([l_1, h_1], [l_2, h_2]) = [l_1 - h_2, h_1 - l_2]$$

Interval Analysis Example (1)

```
// x: [-inf, inf]
if input() {
  // x: [-inf, inf]
  x = 0;
  // x: [0, 0]
} else {
  // x: [-inf, inf]
  x = 2;
  // x: [2, 2]
}
// x: [0, 2]
y = x + 2;
// y: [2, 4]
```

Interval Analysis Example (2)

```
// x: [-inf, inf]
if x <= 0 {
  // x: [-inf, inf]
  y = -x;
  // y: [-inf, inf]
} else {
  // x: [-inf, inf]
  y = x;
  // y: [-inf, inf]
}
// y: [-inf, inf]
```

Control Sensitivity — Transfer Functions

- We can exploit the information available in conditionals

$$t_v : \text{State} \rightarrow \mathcal{P}(\text{State} \times \text{Node})$$

$x = e : \quad t_v(\sigma) = \{(\sigma[x \mapsto \text{eval}(\sigma, e)], \text{succ}(v))\}$
 $\text{if } x \leq n : \quad t_v(\sigma) = \{(\sigma[x \mapsto \text{filter}_{\leq}(\sigma(x), n)], \text{true}(v)),$
 $\quad \quad \quad (\sigma[x \mapsto \text{filter}_{>}(\sigma(x), n)], \text{false}(v))\}$
 $\text{return} : \quad t_v(\sigma) = \emptyset$

Control Sensitivity — Filter Functions

$$filter_{\leq}([l, h], n) = [l, h] \sqcap [-\infty, n]$$

$$filter_{>}([l, h], n) = [l, h] \sqcap [n + 1, \infty]$$

Control Sensitivity — Propagation Algorithm

PropagationWorkListAlgorithm($t_1, \dots, t_n, \sigma_{start}$):

$(x_1, x_2, \dots, x_n) \leftarrow (\sigma_{start}, \perp, \dots, \perp)$

$W \leftarrow \{v_1, \dots, v_n\}$

while $W \neq \emptyset$:

$v_i \leftarrow W.removeOne()$

$Y \leftarrow t_{v_i}(x_i)$

for $(y, v_j) \in Y$:

$z \leftarrow x_j \sqcup y$

if $x_j \neq z$:

$x_j \leftarrow z$

$W.add(v_j)$

return x

Control Sensitivity — Example

```
// x: [-inf, inf]
if x <= 0 {
  // x: [-inf, 0]
  y = -x;
  // y: [0, inf]
} else {
  // x: [1, inf]
  y = x;
  // y: [1, inf]
}
// y: [0, inf]
```

Fixed Point — Tarski's Theorem

Theorem (Tarski¹)

If L is a complete lattice and $f : L \rightarrow L$ is monotone, then f has a least fixed point, given by

$$\text{lfp}(f) = \bigcap \{x \in L \mid f(x) \sqsubseteq x\}$$

- ▶ However, we may not find the lfp within finite iterations
- ▶ If $f(x) \sqsubseteq x$ (x is a pre-fixed point), then $\text{lfp}(f) \sqsubseteq x$

¹A lattice-theoretical fixpoint theorem and its applications (Tarski, 1955)

Fixed Point — Example

```
x = 0;  
// [0, 0]  
while input() {  
    // [0, 0]  
    x = x + 1;  
    // [1, 1]  
}
```

$\Rightarrow$

```
x = 0;  
// [0, 1]  
while input() {  
    // [0, 1]  
    x = x + 1;  
    // [1, 2]  
}
```

...

Widening — Definition

- ▶ Widening² ensures termination even when the lattice has an infinite height
- ▶ Involves a binary operator, $\nabla : L \times L \rightarrow L$, which satisfies:
 1. $\forall x, y \in L. x \sqcup y \sqsubseteq x \nabla y$
 2. $\forall z_0 \sqsubseteq z_1 \sqsubseteq z_2 \sqsubseteq \dots$: the sequence $y_0, y_1, y_2, \dots$ defined by $y_0 = z_0$ and $y_{i+1} = y_i \nabla z_{i+1}$ for $i = 0, 1, \dots$ converges after a finite number of steps

²Abstract interpretation: A unified lattice model for static analysis of programs by construction or approximation of fixpoints (Cousot and Cousot, 1977)

Widening — Approximating the Least Fixed Point

We can approximate $\text{lfp}(f)$ by computing:

$$\begin{aligned}x_0 &= \perp \\x_{i+1} &= x_i \nabla f(x_i)\end{aligned}$$

For some k , we have $x_{k+1} = x_k$, and $\text{lfp}(f) \sqsubseteq x_k$

- ▶ $f(x_k) \sqsubseteq x_k \nabla f(x_k) = x_{k+1} = x_k$
- ▶ ∇ does not need to be symmetric ($x \nabla y \neq y \nabla x$ in general)
- ▶ We combine abstract information from the previous (left-hand) and the current iteration (right-hand)
- ▶ Idea: to coarsen abstract values that are unstable

Widening — Widening Operator

$$\nabla' : \text{Interval} \times \text{Interval} \rightarrow \text{Interval}$$

$$\perp \nabla' y = y$$

$$x \nabla' \perp = x$$

$$[l_1, h_1] \nabla' [l_2, h_2] = [l_3, h_3]$$

where

$$l_3 = \begin{cases} l_1 & \text{if } l_1 \leq l_2 \\ -\infty & \text{otherwise} \end{cases} \quad h_3 = \begin{cases} h_1 & \text{if } h_1 \geq h_2 \\ \infty & \text{otherwise} \end{cases}$$

$$\nabla : \text{State} \times \text{State} \rightarrow \text{State}$$

$$(\sigma_1 \nabla \sigma_2)(x) = \sigma_1(x) \nabla' \sigma_2(x)$$

Widening — Algorithm

PropagationWithWidening($t_1, \dots, t_n, \sigma_{start}$):

$(x_1, x_2, \dots, x_n) \leftarrow (\sigma_{start}, \perp, \dots, \perp)$

$W \leftarrow \{v_1, \dots, v_n\}$

while $W \neq \emptyset$:

$v_i \leftarrow W.removeOne()$

$Y \leftarrow t_{v_i}(x_i)$

for $(y, v_j) \in Y$:

$z \leftarrow x_j \nabla y$

if $x_j \neq z$:

$x_j \leftarrow z$

$W.add(v_j)$

return x

Widening — Example (1)

```
x = 0;  
// [0, 0]  
while input() {  
    // [0, 0]  
    x = x + 1;  
    // [1, 1]  
}  
// [0, 0]
```

$\Rightarrow$

```
x = 0;  
// [0, inf]  
while input() {  
    // [0, inf]  
    x = x + 1;  
    // [1, inf]  
}  
// [0, inf]
```

Widening — Example (2)

```
x = 0;  
// [0, 0]  
while x <= 10 {  
    // [0, 0]  
    x = x + 1;  
    // [1, 1]  
}  
// bot
```

$\Rightarrow$

```
x = 0;  
// [0, inf]  
while x <= 10 {  
    // [0, inf]  
    x = x + 1;  
    // [1, inf]  
}  
// [11, inf]
```

Narrowing — Motivation

- ▶ After widening, narrowing can make the result more precise
- ▶ Where x is the analysis result with widening:

$$\text{lfp}(f) \sqsubseteq f^{i+1}(x) \sqsubseteq f^i(x) \sqsubseteq x$$

- ▶ $\text{lfp}(f) \sqsubseteq f(x) \sqsubseteq x$
- ▶ In general, $x \sqsupseteq f(x) \sqsupseteq f^2(x) \sqsupseteq \dots$ may not converge in finite steps

Narrowing — Definition

We can define a binary operator, $\triangle : L \times L \rightarrow L$, which satisfies:

1. $\forall x, y \in L. y \sqsubseteq x \implies y \sqsubseteq x \triangle y \sqsubseteq x$
2. $\forall z_0 \sqsupseteq z_1 \sqsupseteq z_2 \sqsupseteq \dots$: the sequence $y_0, y_1, y_2, \dots$ defined by $y_0 = z_0$ and $y_{i+1} = y_i \triangle z_{i+1}$ for $i = 0, 1, \dots$ converges after a finite number of steps

Narrowing — Narrowing Operator

$$\Delta' : \text{Interval} \times \text{Interval} \rightarrow \text{Interval}$$

$$\perp \Delta' y = \perp$$

$$x \Delta' \perp = \perp$$

$$[l_1, h_1] \Delta' [l_2, h_2] = [l_3, h_3]$$

where

$$l_3 = \begin{cases} l_2 & \text{if } l_1 = -\infty \\ l_1 & \text{otherwise} \end{cases} \quad h_3 = \begin{cases} h_2 & \text{if } h_1 = \infty \\ h_1 & \text{otherwise} \end{cases}$$

$$\Delta : \text{State} \times \text{State} \rightarrow \text{State}$$

$$(\sigma_1 \Delta \sigma_2)(x) = \sigma_1(x) \Delta' \sigma_2(x)$$

Narrowing — Algorithm

Narrowing($t_1, \dots, t_n, x$):

$W \leftarrow \{v_1, \dots, v_n\}$

while $W \neq \emptyset$:

$v_i \leftarrow W.\text{removeOne}()$

$y \leftarrow \bigsqcup \{\sigma \mid v_j \in \text{dep}^{-1}(v_i) \wedge (\sigma, v_i) \in t_{v_j}(x_j)\}$

$z \leftarrow x_i \triangle y$

if $z \neq x_i$:

$x_i \leftarrow z$

$W \leftarrow W \cup \text{dep}(v_i)$

return x

Narrowing — Example

```
x = 0;  
// [0, inf]  
while x <= 10 {  
    // [0, inf]  
    x = x + 1;  
    // [1, inf]  
}  
// [11, inf]
```

$\Rightarrow$

```
x = 0;  
// [0, 11]  
while x <= 10 {  
    // [0, 10]  
    x = x + 1;  
    // [1, 11]  
}  
// [11, 11]
```

Summary

- ▶ Interval analysis tracks lower and upper bounds of integer variables using a lattice of infinite height
- ▶ Control sensitivity refines analysis precision by exploiting conditional branch information
- ▶ Tarski's theorem guarantees a least fixed point exists, but infinite-height lattices may prevent finite convergence
- ▶ Widening (∇) ensures termination by over-approximating unstable values, at the cost of precision
- ▶ Narrowing ($\triangle$) recovers precision after widening by iteratively tightening the over-approximation